

ROUTE FORECASTING IN HIGH-EFFICIENCY DOMAINS

Group 122

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Problem description and motivation

Maritime traffic monitoring is important for safety, the prevention of illegal activities, and logistics optimization. One of the main challenges is detecting the anomalous behavior in ship trajectories, which may indicate accidents, unauthorized maneuvers or deviations from the expected route.

The work will therefore focus on the implementation of an anomaly detection framework in high-efficiency conditions based on trajectory prediction: starting from the forecasted path of a ship, deviations beyond a set threshold will be identified as anomalies. Emphasis will be placed on computational efficiency, enabling large-scale analysis without compromising accuracy [1]. Our approach will take into consideration that different ship types, like cargo vessels and fishing boats, have different navigation patterns.

The focus will be on a selected maritime region, such as Danish waters, over a defined period using AIS data. Trajectories will be curated for consistency, handling missing or irregularly sampled time series data by interpolation or resampling techniques.

Methodology and implementation plan

A key objective of the project is to achieve high computational efficiency. To this end, maritime space will be discretized into a grid-based representation, going from geographic coordinates to cell indices [2]. This approach allows for efficient probability-based prediction of vessel movements, and enables clear and interpretable visualizations, such as route heatmaps and transition probability maps.

For trajectory prediction, a lightweight recurrent neural network (RNN) will be implemented to model ship positions and forecast short-term movements [3]. If performance is deemed insufficient, a transformer-based model will be implemented as an alternative.

To enhance the learning of the model, data will be sampled at uniform time intervals to address irregularity and sequences will be normalized. Anomalies will be defined as deviations beyond a threshold, tuned based on validation data.

References

- [1] Zhang X. et al. (2022). Vessel Trajectory Prediction in Maritime Transportation: Current Approaches and Beyond
- [2] Nguyen D. et al. (2018). Vessel Trajectory Prediction using Sequence-to-Sequence Models over Spatial Grid
- [3] Capobianco S. et al. (2021). Deep Learning Methods for Vessel Trajectory Prediction Based on Recurrent Neural Networks